

6. En el caso tridimensional tenemos que $\{e_i\}$ define un sistema de coordenadas (de trógrado) y no necesariamente ortogonal, entonces demostrar que:

(a).

$$e^i = \frac{e_j \times e_k}{e_i \cdot (e_j \times e_k)}, \quad i, j, k = 1, 2, 3 \text{ y sus permutaciones cíclicas}$$

Sea $e^i \cdot e_j = \delta_j^i$

$$e^i = c(e_j \times e_k)$$

$$e^i \cdot e_i = c(e_j \times e_k) \cdot e_i = 1$$

$$c \cdot V = 1 \Rightarrow c = \frac{1}{V} \rightarrow V = (e_j \times e_k) \cdot e^i$$

$$\Rightarrow e^i = \frac{e_j \times e_k}{e^i \cdot (e_j \times e_k)}$$

$$(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$$

(b). Si $V = e_1 \cdot (e_2 \times e_3)$ y $\tilde{V} = e^1 \cdot (e^2 \times e^3) \Rightarrow V \tilde{V} = 1$

Sea $\{e_i\}$

$$e^1 = \frac{e_2 \times e_3}{V}, \quad e^2 = \frac{e_3 \times e_1}{V}, \quad e^3 = \frac{e_1 \times e_2}{V}$$

Ahora $\tilde{V} = e^1 \cdot (e^2 \times e^3)$

$$e^2 \times e^3 = \left(\frac{e_3 \times e_1}{V} \right) \times \left(\frac{e_1 \times e_2}{V} \right) = \frac{1}{V^2}$$

$$[(e_3 \times e_1) \times (e_1 \times e_2)]$$

Agora usando:

$$(a \times b) \times (c \times d) = [(a \times b) \cdot d]c - [(a \times b) \cdot c]d$$

$$\begin{cases} a = e_3 \\ b = e_1 \\ c = e_1 \\ d = e_2 \end{cases}$$

$$(e_3 \times e_1) \times (e_1 \times e_2) = [(e_3 \times e_1) \cdot e_2]e_1 -$$

$$[(e_3 \times e_1) \cdot e_1]e_2$$

$$(e_3 \times e_1) \cdot e_2 = e_1 \cdot (e_2 \times e_3) = V$$

$$(e_3 \times e_1) \cdot e_1 = 0$$



Perpendicular a e_1

$$\text{Seu } (e_3 \times e_1) \times (e_1 \times e_2) = V e_1$$

$$e^1 \times e^2 = \frac{1}{V^2} (V e_1) = \frac{e_1}{V}$$

$$\tilde{V} = e^1 \cdot (e^2 \times e^3) = \left(\frac{e_2 \times e_3}{V} \right) \cdot \left(\frac{e_1}{V} \right) = \frac{1}{V^2} (e_2 \times e_3) \cdot e_1$$

$$= \frac{V}{V^2} = \frac{1}{V}$$

$$V \tilde{V} = V \cdot \frac{1}{V} = 1$$

c) Que vetor a satisfaz $a \cdot e^i = 1$?

Demostre que a é único

$$\text{Seu } \{e_i\} \quad a = a^1 e_1 + a^2 e_2 + a^3 e_3$$

$$\Rightarrow a \cdot e^i = a^j (e_j \cdot e^i) = a^j \delta_j^i = a^i$$

$$a \cdot e^i = 1 \quad \forall i \Rightarrow a^i = 1 \quad \forall i$$

$$a = e_1 + e_2 + e_3$$

Si existiera otro vector $\vec{E}$ tal que $E \cdot e^i = 1 \quad \forall i$ entonces sus componentes contravariantes serían $E^i = 1$ igual que las de a . Por lo tanto $b = a$

(d). Dada la base:

$$w_1 = 4i + 2j + k, \quad w_2 = 3i + 3j, \quad w_3 = 2k$$

Encontrar:

i. LAS Bases reciprocas $\{e^i\}$

ii. Las componentes covariantes y contra-variantes del vector $a = i + 2j + 3k$

[I] $e_1 = (4, 2, 1), \quad e_2 = (3, 3, 0), \quad e_3 = (0, 0, 2)$

$$V = e_2 \cdot (e_1 \times e_3)$$

$$e_2 \times e_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 3 & 0 \\ 0 & 0 & 2 \end{vmatrix} = (6, -6, 0)$$

$$V = e_2 \cdot (e_2 \times e_3) = (4, 2, 1) \cdot (6, -6, 0) = 24 - 12 + 0 = 12$$

Usamos $e^i = \frac{e_j \times e_k}{V} \Rightarrow$ permutaciones cíclicas

$$e^1 = \frac{e_2 \times e_3}{V} = \frac{(6, -6, 0)}{12} = \left(\frac{1}{2}, -\frac{1}{2}, 0\right)$$

$$e^3 = \frac{e_1 \times e_2}{\sqrt{\quad}}$$

$$e_1 \times e_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 2 & 1 \\ 3 & 3 & 0 \end{vmatrix} = (-3, +3, 6)$$

$$e^3 = \frac{(-3, 3, 6)}{12} = \left(-\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right)$$

$$e^2 = \frac{e_3 \times e_1}{\sqrt{\quad}}$$

$$e_3 \times e_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 2 \\ 4 & 2 & 1 \end{vmatrix} = (-4, 8, 0)$$

$$e^2 = \frac{(-4, 8, 0)}{12} = \left(-\frac{1}{3}, \frac{2}{3}, 0\right)$$

Base reciprocals:

$$e_1 = \left(\frac{1}{2}, -\frac{1}{2}, 0\right) \quad e^2 = \left(-\frac{1}{3}, \frac{2}{3}, 0\right) \quad e^3 = \left(-\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right)$$

ii).

$$a = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$a = (1, 2, 3)$$

$$a^i = a \cdot e_i$$

$$a^1 = a \cdot e_1 = (1, 2, 3) \cdot (4, 2, 1) = 11$$

$$a^2 = a \cdot e_2 = (1, 2, 3) \cdot (3, 7, 0) = 9$$

$$a^3 = a \cdot e_3 = (1, 2, 3) \cdot (0, 0, 2) = 6$$

$$a_i = a \cdot e^i$$

$$a_1 = a \cdot e^1 = (1, 2, 3) \cdot \left(\frac{1}{2}, -\frac{1}{2}, 0\right) = \frac{1}{2} - 1 + 0 = -\frac{1}{2}$$

$$a_2 = a \cdot e^2 = (1, 2, 3) \cdot \left(-\frac{1}{3}, \frac{2}{3}, 0\right) = -\frac{1}{3} + \frac{4}{3} + 0 = 1$$

$$a_3 = a \cdot e^3 = (1, 2, 3) \cdot \left(-\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right) = -\frac{1}{4} + \frac{1}{2} + \frac{3}{2} = \frac{7}{4}$$

Contravariantes: $a^1 = 11$, $a^2 = 9$, $a^3 = 6$

Covariantes: $a_1 = -\frac{1}{2}$, $a_2 = 1$, $a_3 = \frac{7}{4}$

7. Consideremos las matrices hermitas 2×2 y def. $P.I = \langle a | b \rangle \Leftrightarrow \text{Tr}(A^T B)$
 Comprobemos también la matriz unitaria y las matrices de Pauli.
 Encuentre entonces la base dual asociada a la base de Pauli
 y adicionalmente dado un vector genérico en este espacio vectorial
 encuentre su 1-Forma asociada

• Consideremos el espacio vectorial de matrices hermitas $2 \times 2 \in \mathbb{R}$

• El producto interno está definido por $\langle A | B \rangle = \text{Tr}(A^T B)$

Base dada $\{P\} = \{I, \sigma_1, \sigma_2, \sigma_3\}$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$G_{ij} = \langle \sigma_i | \sigma_j \rangle = \text{Tr}(\sigma_i^T \sigma_j)$$

Entonces:

$$\sigma_0^\dagger = \sigma_0$$

$$\sigma_1^\dagger = \sigma_1$$

$$\sigma_2^\dagger = -\sigma_2 \text{ (Ya que es antisimetrica)}$$

$$\sigma_3^\dagger = \sigma_3$$

$$1. \langle \sigma_0 | \sigma_0 \rangle = \text{Tr}(\sigma_0^\dagger \sigma_0) = \text{Tr}(I \cdot I) = \text{Tr}(I) = 2$$

$$2. \langle \sigma_0 | \sigma_1 \rangle = \text{Tr}(\sigma_0^\dagger \sigma_1) = \text{Tr}(I \cdot \sigma_1) = \text{Tr}(\sigma_1) = 0$$

$$3. \langle \sigma_0 | \sigma_2 \rangle = \text{Tr}(I \cdot \sigma_2) = \text{Tr}(\sigma_2) = 0$$

$$4. \langle \sigma_0 | \sigma_3 \rangle = \text{Tr}(I \cdot \sigma_3) = \text{Tr}(\sigma_3) = 0$$

$$5. \langle \sigma_1 | \sigma_1 \rangle = \text{Tr}(\sigma_1^\dagger \sigma_1) = \text{Tr}(\sigma_1 \cdot \sigma_1) = \text{Tr}(I) = 2$$

$$6. \langle \sigma_1 | \sigma_2 \rangle = \text{Tr}(\sigma_1^\dagger \sigma_2) = \text{Tr}(\sigma_1 \cdot \sigma_2) = \text{Tr}(\sigma_3) = 0$$

$$7. \langle \sigma_1 | \sigma_3 \rangle = \text{Tr}(\sigma_1^\dagger \sigma_3) = \text{Tr}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) = 0$$

$$8. \langle \sigma_2 | \sigma_2 \rangle = \text{Tr}(\sigma_2^\dagger \sigma_2) = \text{Tr}(-\sigma_2 \cdot \sigma_2) = \text{Tr}(I) = 2$$

Cuidadosamente vemos que $\sigma_2^\dagger = -\sigma_2$

$$\Rightarrow \sigma_2 \sigma_2 = -\sigma_2 \sigma_2 = I$$

$$\text{Por eso } \text{Tr}(-I) = -2$$

$$9. \langle \sigma_2 | \sigma_3 \rangle = \text{Tr}(\sigma_2^\dagger \sigma_3) = \text{Tr}(-\sigma_2 \sigma_3) = \text{Tr}(-i\sigma_1) = 0$$

$$10. \langle \sigma_3 | \sigma_3 \rangle = \text{Tr}(\sigma_3^\dagger \sigma_3) = \text{Tr}(\sigma_3 \sigma_3) = \text{Tr}(I) = 2$$

$$G = \begin{pmatrix} \langle \sigma_0, \sigma_0 \rangle & \langle \sigma_0, \sigma_1 \rangle & \langle \sigma_0, \sigma_2 \rangle & \langle \sigma_0, \sigma_3 \rangle \\ \langle \sigma_1, \sigma_0 \rangle & \langle \sigma_1, \sigma_1 \rangle & \langle \sigma_1, \sigma_2 \rangle & \langle \sigma_1, \sigma_3 \rangle \\ \langle \sigma_2, \sigma_0 \rangle & \langle \sigma_2, \sigma_1 \rangle & \langle \sigma_2, \sigma_2 \rangle & \langle \sigma_2, \sigma_3 \rangle \\ \langle \sigma_3, \sigma_0 \rangle & \langle \sigma_3, \sigma_1 \rangle & \langle \sigma_3, \sigma_2 \rangle & \langle \sigma_3, \sigma_3 \rangle \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Ahora encontramos el dual de $\{P\}$

$$\langle \sigma^\mu, \sigma_\nu \rangle = \delta^\mu_\nu$$

$$\sigma^\mu = \sum_\alpha (G^{-1})^{\mu\alpha} \sigma_\alpha$$

G^{-1} es la inversa de G

$$G = \text{diag}(2, 2, -2, 2) \Rightarrow G^{-1} = \text{diag}\left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right)$$

$$\begin{aligned} \sigma^0 &= (G^{-1})^{00} \sigma_0 + (G^{-1})^{01} \sigma_1 + (G^{-1})^{02} \sigma_2 + (G^{-1})^{03} \sigma_3 \\ &= \frac{1}{2} \sigma_0 \end{aligned}$$

$$\sigma^1 = \frac{1}{2} \sigma_1$$

$$\sigma^2 = -\frac{1}{2} \sigma_2$$

$$\sigma^3 = \frac{1}{2} \sigma_3$$

$$\text{Seu} = \sigma_0 = \frac{1}{2} \sigma_0, \sigma_1^1 = \frac{1}{2} \sigma_1, \sigma^2 = -\frac{1}{2} \sigma_2, \sigma_3 = \frac{1}{2} \sigma_3$$

Verificamos: $\langle \sigma_0^0 \sigma_0 \rangle = \langle \frac{1}{2} \sigma_0, \sigma_0 \rangle = \frac{1}{2} \cdot 2 = 1$ e $\langle \sigma_0^0, \sigma_1 \rangle = 0$.

Vector genérico $|v\rangle \in S$ é a uma matriz hermitica

$$A = a_0 \sigma_0 + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3$$

$$w_A(B) = \langle A | B \rangle$$

$$w_A = \sum_{\mu} w_A(\sigma_{\mu}) \sigma^{\mu}$$

$$w_A(\sigma_{\mu}) = \langle A | \sigma_{\mu} \rangle$$

$$\langle A | \sigma_{\mu} \rangle = \text{Tr}(A^T \sigma_{\mu})$$

$$A^T = a_0 \sigma_0^T + a_1 \sigma_1^T + a_2 \sigma_2^T + a_3 \sigma_3^T = a_0 \sigma_0 + a_1 \sigma_1 - a_2 \sigma_2 + a_3 \sigma_3$$

$$\langle A | \sigma_{\mu} \rangle = \text{Tr}(A^T \sigma_{\mu})$$

$E_j: \mu=0$

$$\langle A | \sigma_0 \rangle = \text{Tr}(A^T \sigma_0) = \text{Tr}(A^T) = \text{Tr}(a_0 \sigma_0 + \sigma_1 \sigma_1 - a_2 \sigma_2 + \sigma_3 \sigma_3) = 2a_0$$

$$\text{Tr}(\sigma_0) = 2, \text{Tr}(\sigma_1) = 0, \text{Tr}(\sigma_2) = 0, \text{Tr}(\sigma_3) = 0$$

$$\langle A | \sigma_i \rangle = \text{Tr}(A^T \sigma_i) = \text{Tr}((a_0 \sigma_0 + a_1 \sigma_1 - a_2 \sigma_2 + a_3 \sigma_3) \sigma_i)$$

$$\sigma_0 \sigma_1 = \sigma_1, \sigma_1 \sigma_1 = \bar{I}, \sigma_2 \sigma_1 = -i \sigma_3, \sigma_3 \sigma_1 = \sigma_2$$

$$A^T \sigma_1 = a_0 \sigma_1 + a_1 \bar{I} - a_2 (-i \sigma_3) + a_3 \sigma_2 = a_0 \sigma_1 + a_1 \bar{I} + i a_2 \sigma_3 + a_3 \sigma_2$$

$$\text{Traza} = \text{Tr}(a_1 \bar{I}) = 2a_1$$

$$\langle A | \sigma_0 \rangle = 2a_0, \langle A | \sigma_1 \rangle = 2a_1, \langle A | \sigma_2 \rangle = -2a_2, \langle A | \sigma_3 \rangle = 2a_3$$

$$w_A(\sigma_0) = 2a_0, w_A(\sigma_1) = 2a_1, w_A(\sigma_2) = -2a_2, w_A(\sigma_3) = 2a_3$$

$$1. \text{ Formula} = w_A = [2a_0] \sigma^0 + [2a_1] \sigma^1 + [-2a_2] \sigma^2 + [2a_3] \sigma^3$$

Sushkuyembi:

$$\sigma_0^1 = \frac{1}{2} \sigma_0, \sigma_1^1 = \frac{1}{2} \sigma_1, \sigma_2^1 = -\frac{1}{2} \sigma_2, \sigma_3^1 = \frac{1}{2} \sigma_3$$

$$\text{Terimoy, } w_A = a_0 \sigma_0 + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3$$

$$\text{Baxe dual: } \sigma_0^1 = \frac{1}{2} \sigma_0, \sigma_1^1 = \frac{1}{2} \sigma_1, \sigma_2^1 = -\frac{1}{2} \sigma_2, \sigma_3^1 = \frac{1}{2} \sigma_3$$

$$A = a_0 \sigma_0 + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3$$

$$w_A = a_0 \sigma_0 + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3$$